

LAKSHYA KUMAR

Engineering Leader | AI, Web3, Distributed Systems | 7+ Years

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PROFESSIONAL SUMMARY

Engineering leader with 7+ years across fintech, AI, and Web3. I've shipped a Polygon and Ethereum payments platform doing 25,000+ transactions a day, an institutional fund tokenization platform on EVM and Provenance, and an enterprise RAG assistant that's now in production. These days I'm tech lead at Payram, building ZepaPay, an on-ramp/off-ramp crypto payments platform, where I also own the AI agents behind support and operations. I move between architecture review and Go concurrency code without much friction, and I've grown engineering teams from five to fifteen while staying hands-on. Core maintainer of Hyperledger Bevel, author of Mjolnir (an open-source production-readiness audit), and a regular technical writer on Medium.

CORE SKILLS

Languages: TypeScript, JavaScript, Python, Go, Solidity, Rust

Backend & Systems: Node.js, FastAPI, microservices, REST, event-driven queues, concurrent worker pools, distributed databases, ACID and BASE design, connection pooling, horizontal scaling, Postgres, Redis

AI & Agents: LangChain, LangGraph, CrewAI, PydanticAI, RAG pipelines, hybrid retrieval (BM25 + dense, RRF), reranking, semantic search, vector databases, MCP servers, tool orchestration, agent guardrails, prompt-injection defense, model routing, AI memory

Evals, Fine-tuning & LLMops: LLM-as-judge, trajectory evals, golden datasets, CI eval gates, drift detection, fine-tuning (LoRA / QLoRA, PEFT), model serving (vLLM, Triton, TensorRT-LLM), quantization (AWQ/GPTQ/FP8), MLflow, Langfuse, p95/p99 SLOs

Production-Readiness: OpenTelemetry, structured logging, RED metrics, tail-latency budgets, circuit breakers, distributed rate limiting, leader election, graceful shutdown, distroless containers, fuzz and property-based testing, schema migrations

Blockchain: EVM, Provenance, Hyperledger Fabric and Besu, Polygon, Avalanche, ERC-1400, ERC-4337, account abstraction, cross-chain bridges, tokenization

Cloud & DevOps: AWS (ECS, EC2, Lambda, S3, ECR, CodePipeline), Docker, Kubernetes, GPU orchestration, autoscaling, GitHub Actions, Jenkins, GitOps, blue-green / canary / rolling deployments

Leadership: architecture, team lead (up to 15 engineers), mentorship, hiring, delivery management, technical strategy

PROFESSIONAL EXPERIENCE

Tech Lead (Contract) | Payram

Nov 2025 - Present

- Tech lead for a self-hosted crypto payments platform written in Go, owning architecture, implementation, and the production rollout.
- Tech lead on ZepaPay, an on-ramp/off-ramp crypto payments platform built for Consult Now on Payram's rails. It moves money between fiat and crypto, handles KYC and compliance, and runs settlement end to end.
- Built and shipped the AI agents in the payments and support flows. They call tools against strict schemas, sit behind input, output, and scope guardrails to shut down prompt injection, and route between models to keep cost and latency down. Nothing ships until it clears an eval suite of trajectory checks plus an LLM judge.
- Built event-driven push and pull queues with concurrent workers and cron-scheduled async settlement, decoupling payment initiation from dispatch. Worker pools scale horizontally and blue-green pipelines give zero-downtime releases with sub-second dispatch under concurrent load.
- Wired in OpenTelemetry across the HTTP, Postgres, and Redis paths with W3C trace-context propagation; structured JSON logs on stdout carry request and trace IDs. Cut mean detection time on the last two production incidents to under 4 minutes.
- Every outbound call (DB, Redis, third-party rails) goes through a circuit breaker that classifies infrastructure errors separately from client-fault errors. A partial Redis blip in March never reached the API tier.
- Postgres advisory locks gate the cron-scheduled settlement worker so exactly one replica runs per tick, which closed out a duplicate-charge race we hit in the first week of the rollout.
- Redis-backed token-bucket rate limiter with an explicit fail-closed branch on Redis errors; holds the 200 rps cap across replicas, verified at 1,200 rps with 6 instances during load runs.
- Distroless multistage image running as nonroot, ~26 MB final size (down from ~410 MB on the first cut). BuildKit cache mounts pulled CI build time from roughly 8 minutes down to about 110 seconds.
- Graceful SIGTERM handler with a bounded 30-second drain and reverse-order resource teardown; the connection-pool panics we used to see in deploy logs are gone.

VP, Web3 and AI Solutions | Formidium

Feb 2024 - Oct 2025

- Grew the Web3 and AI organization from six to fifteen engineers across two product lines, owning hiring, architecture review, and quarterly delivery.
- Delivered an institutional fund tokenization platform on EVM and Provenance using ERC-1400, with ACID semantics on the off-chain ledger and BASE for distributed on-chain state.
- Built an enterprise RAG assistant in Python (LangChain, LangGraph, Next.js front-end) that cut analyst research time by 60%. Retrieval combined keyword and vector search with a reranking pass, and every answer carried citations back to source so analysts could check them.
- Put the assistant behind an eval suite so quality regressions got caught in CI before release, scoring answers against a golden set with an LLM judge. Kept p95 latency in check with caching and streamed responses so it felt fast.
- Brought NAV update lag from minutes down to seconds through blockchain oracles and an event-driven update pipeline.
- Removed single points of failure in custody by re-architecting the wallet layer around multi-sharded key storage and AWS Secrets Manager.

- Standardised the Web3 services on a layered architecture with one-way dependencies, JSON logging on stdout, and OpenTelemetry tracing across HTTP, Postgres, and the wallet RPC. Median incident triage moved from about 45 minutes to roughly 12 over two quarters.
- Moved Postgres schema changes to versioned, ordered migration files run as a deploy pre-hook. The “new binary on old schema” incident class went from roughly one a month to zero across the year that followed.
- Built a Redis-backed distributed rate limiter for the RAG assistant's document API with a fail-open branch on Redis blips for the read-mostly workload; held caps across the 6-replica analyst pilot at peak (~480 rps).
- Released MCP-compatible AI tooling for internal knowledge retrieval and developer workflows; rolled out SSO across Formidium products, and designed the monitoring, alerting, failover, and connection pooling for the high-throughput services.

Backend Engineer | Xalts

Apr 2023 - Dec 2023

- Owned all backend and blockchain modules over a six-month delivery window while mentoring a team of five.
- Onboarded banking clients to a private EVM chain on Avalanche and Hyperledger Besu via a custodial wallet with Web2 onboarding and automated key generation.
- Used ERC-4337 account abstraction to enable programmable enterprise transaction flows through smart contract wallets.
- Implemented cross-chain transfers between Besu and Avalanche with a blockchain indexer and token bridge, cutting settlement from hours to under a minute.
- Deployed services on AWS (VMs, ECS, S3) using blue-green and rolling rollouts, API gateway rate limiting, and cross-AZ load balancing.
- Split the test suite into a fast tier of about 340 unit tests on in-memory fakes (finishing in under 10 seconds) and an integration tier on testcontainers for Postgres and the Besu node, both behind the same make test target the team ran locally.
- Added a profiling endpoint behind a private listener on the bridge service and caught a heap leak in the relayer about 20 minutes after the symptom showed up in production.

Blockchain Lead | Quadrant.io

Dec 2021 - Mar 2023

- Led a five-engineer team across multiple blockchain applications and production releases.
- Designed and built a Polygon and Ethereum platform running 25,000+ daily transactions with zero downtime, using concurrent processing, message queues, Go channels, and retry-and-recovery logic.
- Designed the platform's private EVM chain for multi-region data consistency, including distributed database architecture, full-node infrastructure, bridge, explorer, and APIs.
- Designed tokenomics and an NFT-based incentive system for the eQUAD token; deployed services on AWS Lambda for elastic scale.
- Wrote fuzz targets for the address validator and the bridge encoder, with crash corpora committed back to the repo as regression tests. One run uncovered a UTF-8 normalisation bug that had been live in staging for over a month.
- Benchmarked the hot-path RLP encoder and caught a refactor in review that would have shipped about 3.4x slower, before the change went out.

Concept Engineer | Everledger

Feb 2021 - Oct 2021

- Shipped React provenance widgets for diamond authenticity verification into client-facing production applications.
- Expanded the platform into the Chinese market through a WeChat API integration and Hyperledger Fabric extensions for local compliance.
- Delivered a Fabric-to-Besu migration proof of concept that informed the company's later chain strategy.

Software Engineer | Accenture

Sep 2018 - Jan 2021

- Core contributor to Hyperledger Bevel, the open-source Kubernetes framework for production blockchain deployment now used widely across enterprise.
- Built deployment pipelines for Hyperledger Fabric, Hyperledger Besu, and JP Morgan Quorum with CI/CD and GitOps via Jenkins and Travis CI.
- Contributed internal blockchain solutions back to Hyperledger Labs and the wider open-source ecosystem.

SELECTED PROJECTS

ZepaPay | zepapay.com

On-ramp/off-ramp crypto payments platform built on Payram's rails for Consult Now. Handles fiat-to-crypto conversion, KYC and compliance, and settlement. I was tech lead, and also built the AI agents that run behind support and operations.

Capstok | capstok.com

Hands-on learning platform for builders, with 7,500+ challenges across 14 tracks covering AI/ML, blockchain, cloud, security, and systems. It plugs into AI assistants: Claude, Cursor, or any MCP-aware agent can pull the material over a read-only API to teach and quiz you, in 40+ languages.

Mjolnir | github.com/lakshyakumar/mjolnir

Open-source production-readiness audit, language and agent agnostic. Walks any backend service against a 19-item canonical checklist (circuit breakers, distributed rate limiting, leader election, OpenTelemetry tracing, RED metrics, graceful shutdown, distroless images, schema migrations, and more) and returns a single WORTHY or NOT WORTHY verdict with file-level evidence. Ships as a skill for Claude Code and Cowork.

Hyperledger Bevel | github.com/hyperledger/bevel

Open-source accelerator for distributed ledger deployment on Kubernetes.

Advanced RAG Assistant | github.com/lakshyakumar/Advanced-RAG

Production AI assistant for document Q&A; rephrases and simplifies queries before retrieval, with 60% faster research through LangChain and LangGraph orchestration.

RepoMind | github.com/lakshyakumar/repomind

Repository-intelligence MCP server for coding agents. Exposes structured codebase context, semantic search, and file-level queries as AI-callable tools.

TypeScript MCP Server | github.com/lakshyakumar/typescript-mcp

Production-grade MCP server in TypeScript over JSON-RPC 2.0; modular and horizontally scalable.

Filmy Agent | github.com/lakshyakumar/filmy_agent

Rust agent that generates cinematic prompts for image and video models.

Fund Tokenization Platform

Staking, sales, and token contracts across EVM and Provenance for institutional funds, using ERC-1400 security tokens.

WRITING & OPEN SOURCE

Technical blog: medium.com/@lklsquare. Topics include account abstraction (ERC-4337), AI agent tooling, RAG systems, vector search, and MCP servers.

- Mastering GitFlow: A Production-Ready Branching Strategy for Enterprise Development Teams
- Teaching AI to Remember: Episodic, Semantic, and Procedural Memory in AI
- Vector Search in MongoDB: Best Practices for Real-Time Semantic Retrieval

Open source: core contributor to Hyperledger Bevel; author of Mjolnir (production-readiness audit skill); maintainer of Turbulent (Node performance toolkit), Wallact (ethers.js contract-interaction wrapper), and XIToDb (Excel-to-database loader) on npm.

CERTIFICATIONS

- AWS Certified Cloud Practitioner (Amazon Web Services)
- Building AI-Powered Search with MongoDB Vector Search (MongoDB, 2025)
- Building RAG Apps Using MongoDB (MongoDB, 2025)

EDUCATION

M.Tech, Computational Mathematics | NIT Surathkal, Karnataka *Jul 2016 - Jun 2018*

B.Tech | Delhi Technological University *Aug 2011 - Jun 2015*